

## Project Proposal

# Sentiment Analysis and Rating Prediction on Hotel Reviews

6CS012

Artificial Intelligence and Machine Learning

Herald College Kathmandu

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**Table Of Content**

**1. Research Question..... 3**

**2. Title.....3**

**3. Dataset Description.....3**

    Source..... 3

    Dataset Composition.....4

    Collection And Preparation Method..... 5

    Challenges.....5

**4. Model Pipeline.....5**

**5. Preliminary Data Analysis.....6**

    Class distribution..... 6

    Sample Visualization..... 7

## **1. Research Question**

Can a Recurrent Neural Network (RNN) or a Transformer-based model accurately predict numerical hotel ratings (1–5) by analyzing the semantic content of customer reviews from TripAdvisor?

## **2. Title**

A Performance Analysis of Deep Learning and Transformer Architectures for Multiclass Sentiment Classification of Hotel Reviews.

## **3. Dataset Description**

### **Source**

The dataset consists of 20,491 hotel reviews extracted from TripAdvisor, specifically curated for sentiment analysis and rating prediction tasks.

## Dataset Composition

**Number of Classes:** 5 (Ratings 1, 2, 3, 4, and 5)

**Total Number of Reviews:** 20,491

**Data Split:** The data will be divided into Training, Validation, and Test sets (e.g., 70% Train, 15% Val, 15% Test)

### Features:

- **Review:** Textual feedback provided by guests
- **Rating:** Numerical target variable (1–5)

## Collection And Preparation Method

As the dataset is sourced from TripAdvisor, the following preparation steps will be applied:

**Text Preprocessing:** Conversion to lowercase, removal of punctuation, special characters, and stopwords to reduce noise.

**Tokenization and Embedding:** Transforming text into numerical sequences using techniques such as Word2Vec, or RNN model embedding layer.

**Padding and Truncation:** Ensuring all input sequences have a uniform length for model processing.

**Data Splitting:** Implementing a stratified split to ensure each rating class is represented proportionally in the training, validation, and test sets.

## Challenges

**Class Imbalance:** Reviews often lean toward very positive (5) or very negative (1) ratings, which may bias the model against middle-range ratings (2–4).

**Variable Review Length:** Long, detailed reviews compared to very short ones require effective sequence handling (like attention mechanisms) to capture relevant sentiment.

## 4. Model Pipeline

We will be building three models to evaluate performance:

- **Baseline Model:** A simple Recurrent Neural Network (RNN) with a trainable Embedding layer.
- **Deeper Architecture with Regularization:** A Long-Short Term Memory (LSTM) with a trainable Embedding layer.
- **Fine-Tuning a Pre-Trained Model:** A Long-Short Term Memory (LSTM) with pre-trained Word2Vec embeddings.

## 5. Preliminary Data Analysis

### Class distribution

Visualizing the frequency of each rating (1 through 5) to identify potential imbalances.

Rating Counts:

Rating

1 1421

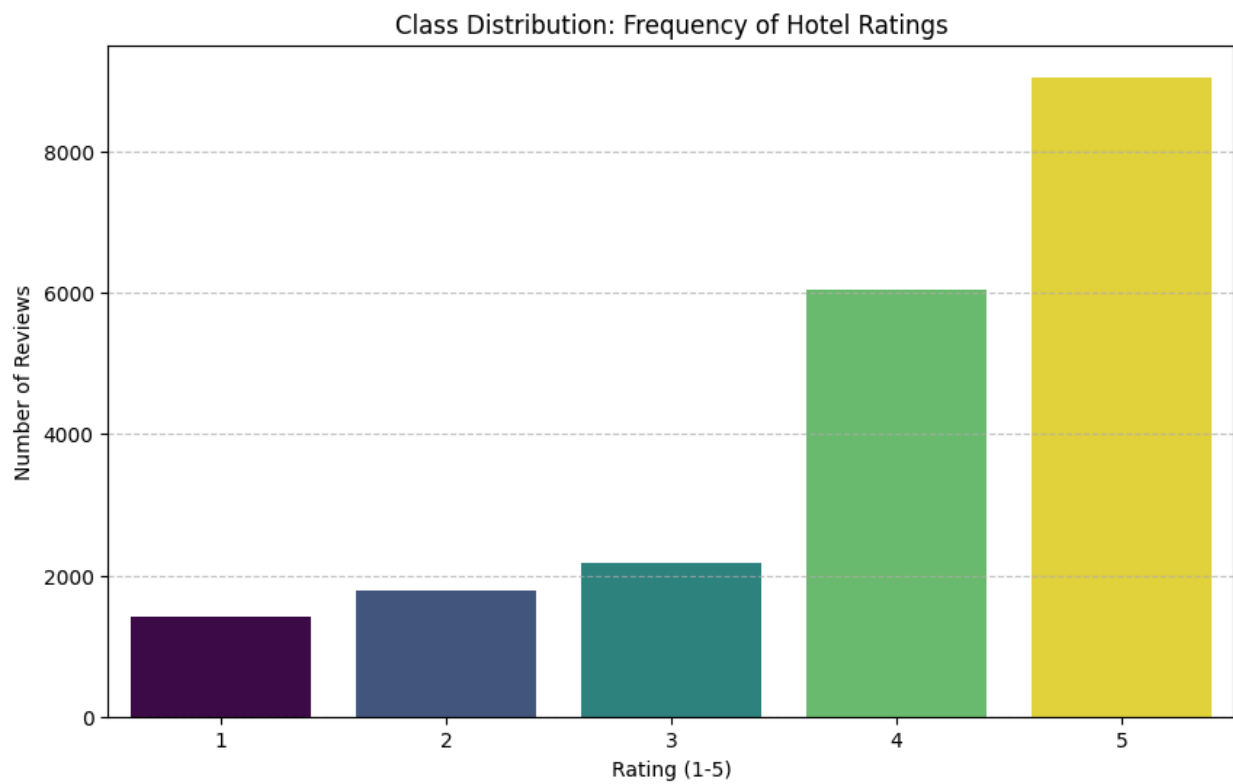
2 1793

3 2184

4 6039

5 9054

Name: count, dtype: int64



## Sample Visualization

We will display sample reviews for each rating class to understand the linguistic differences between a "1-star" and a "5-star" review. For example:

```
-----  
SAMPLE REVIEWS PER RATING  
-----
```

```
--- Rating 1 ---
```

```
Sample: horrible customer service hotel stay february 3rd 4th 2007my friend picked hotel monaco appealing website onl
```

```
--- Rating 2 ---
```

```
Sample: ok nothing special charge diamond member hilton decided chain shot 20th anniversary seattle, start booked sui
```

```
--- Rating 3 ---
```

```
Sample: nice rooms not 4* experience hotel monaco seattle good hotel n't 4* level.positives large bathroom mediterrane
```

```
--- Rating 4 ---
```

```
Sample: nice hotel expensive parking got good deal stay hotel anniversary, arrived late evening took advice previous t
```

```
--- Rating 5 ---
```

```
Sample: unique, great stay, wonderful time hotel monaco, location excellent short stroll main downtown shopping area,
```